ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

- **English Name: Benjamin ESCUDER**
- **Chinese Name:**
- **Email: escuder.benjamin@gmail.com**
- **Student ID: J153040001**
- **University: National Sun Yat-sen University**

Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer: In this sci.med category there are 594 documents. Isolating specific categories helps find more meaningful patterns within these categories.

AI used for this answer: No

If yes, how:

External references (links):

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Using dropna() would delete 40% of the dataset and we will lose too much information and introduce bias if the missing labels are not random. An objective alternative would be to train a model on the labeled documents to predict the missing column value.

AI used for this answer: No

If yes, how:

External references (links):

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dropna.html (to see if there were a way to proportionally delete across all the categories with dropna().)

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer: In the original (2257 docs), there were 480 alt.atheism (21.3%), 584 comp.graphics (25.9%), 594 sci.med (26.3%) and 599 soc.religion.christian (26.5%). While in the sampled distribution: alt.atheism 214 (21.4%), comp.graphics 278 (27.8%), sci.med 260 (26.0%), soc.religion.christian 248 (24.8%), so there's quite a difference especially on alt.atheism and soc.religion.christian.

The risk is that the simple random sampling will proportionally more impact the minority classes, such as alt.atheism here. It can underestimate rare classes in a imbalanced or in a small dataset and even make them disappear.

AI used for this answer: No

If yes, how:

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer: Since we are now taking into account words that were before stopwords, there are more features. There are 35788 features while there were 35482 before. The overall sparsity is now lower because stopwords appear in nearly all documents, so their features aren't zero and participating in making the matrix less sparse.

AI used for this answer: No

If yes, how:

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

The size of the vocabulary shrinks fast as we set the variance threshold higher. The words that have a variance close to zero are the first to get eliminated. There are misspelled words and identification numbers that appear once. It is safe to remove them because they do not carry any information because it never varies, its variance is null.

AI used for this answer: No

If yes, how:

External references (links):

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer: Pearson's top 10 is mixed with domain terms (vga, video, card) but has also noisy generic words (thanks, hi, looking) while Spearman's top 10 has more technical words but still mixed with generic stop words.

As written in the notebook, Pearson uses a linear relationship between term count and the class indicator, so it is really sensitive to outliers while Spearman uses ranks and assesses monotonic relationships (whether linear or not).

The rank order of a word whose count is high in a few long docs will still match the class so the word will have a high Spearman correlation, but the high value distortion will interfere with the linear fit resulting in a low Pearson correlation.

AI used for this answer: No

If yes, how:

External references (links):

https://en.wikipedia.org/wiki/Spearman's%20rank%20correlation%20coefficient

---

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Lowering minSup from 9 to 3 allows to have more frequent patterns passing the threshold so having a larger output (710 against 9547), but the search is then longer, and it takes a lot more time (0.15s against 2.64s).

Answer (2) — Top-K k vs MaxFPGrowth minSup:

MaxFPGrowth minSup uses a threshold while the Top-K fixes the output size to K patterns.

Answer (3) — Maximal vs standard frequent patterns:

A maximal frequent pattern contains other frequent patterns in it, so we do not need to list all those smallest patterns, keeping only the maximal one, saving space.

AI used for this answer: Yes

If yes, how: To correct my English grammatical errors

External references (links):

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer: X_counts has 35788 dimensions, and we cannot plot them on a 2 or either a 3 axes graph. So, we have to choose what dimensions to plot. We must use dimension reduction and tools like PCA, t-SNE or UMAP, as stated in the notebook, to compress the data into 2 or 3 components to plot it.

AI used for this answer: No

If yes, how:

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer: UMAP is showing separate cluster islands for the documents categories (and t-SNE less so). PCA doesn't show any separation, with a huge overlap of categories around a single point, and this is because it is linear for finding the direction of the maximum variance while text categories are not linearly separable. T-SNE and UMAP use local neighborhoods. But t-SNE does not preserve global distances, only local ones, hence, the gap between two clusters doesn't really help to measure how different the groups are.

AI used for this answer: Yes

If yes, how: For correcting my draft

External references (links):

http://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html

https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html

https://umap-learn.readthedocs.io/en/latest/basic_usage.html

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

4 binary columns are created, one per category. If we give the raw integer column as input, it is false to treat them as ordered numbers, assuming that one category is greater than another, and the categories are nominal and don't have any real order. One-hot binarization eliminates that by giving each category its own column, so each category is treated equally distant from any other ones.

AI used for this answer: No

If yes, how:

External references (links):
